

PRE-SEED GOVERNMENT VENTURE CAPITAL FUNDS*

Journal of International Entrepreneurship, forthcoming

Douglas Cumming

Associate Professor and Ontario Research Chair
York University - Schulich School of Business
4700 Keele Street
Toronto, Ontario M3J 1P3
Canada

<http://www.schulich.yorku.ca/>
<http://Douglas.Cumming.com/>
Douglas@Cumming.com

Sofia Johan

University of Tilburg
AFM Senior Research Fellow, Tilburg Law and Economics Centre (TILEC)
Postbus 90153
5000 LE Tilburg
The Netherlands

<http://ssrn.com/author=370203>
S.A.Johan@uvt.nl

This draft: 1 November 2007

* We owe special thanks to the Department of Industry, Tourism and Resources of the Corporate Strategy Branch of the Government of Australia for helpful comments and inspiring and sponsoring this research, and to the Australian Venture Capital Association for their helpful support.

PRE-SEED GOVERNMENT VENTURE CAPITAL FUNDS

Abstract

This paper analyses a Pre-Seed Fund (“PSF”) government venture capital (“VC”) program for the purpose of improving our understanding about effective public policy towards entrepreneurial finance. The PSF program is a public-private partnership started in 2002 for the purpose of fostering more investment in nascent high-tech entrepreneurial companies in Australia. Data from Venture Economics indicate PSFs are the primary provider of seed stage VC in Australia, but PSFs are not more likely to invest in high-tech companies than other types of VC funds. PSFs have smaller portfolios (number of investees) per manager than other types of VC funds, and are more likely to invest in companies resident in the same state, but do not stage and syndicate more frequently than other types of VC funds. Overall, therefore, the structure of the program has given rise to mixed performance in terms of finance and governance provided to nascent high-tech entrepreneurial companies. As well, there is also suggestive evidence that the PSF program diminishes the incentives for Innovation Investment Funds (a previously existing Australian government VC fund program) to invest in seed stage ventures, and hence competing government initiatives appear to be crowding out one another. Further evidence suggests that among the four PSFs in existence, one PSF has outperformed the other PSFs in regards to the investee company patents and financial statement performance, even though this fund has invested less money and charged lower management fees than its counterparts. Hence, a further lesson from the PSF program is that the impact of government sponsored VC funds depends not only on the design of the program but also on the selection of the VC managers carrying out the investments.

Keywords: Venture Capital; Government; Public Policy; Entrepreneurship

JEL Classification: G24, G28, G31, G32, G35

1. Introduction

The importance of venture capitalists (“VCs”) in providing funds for small and medium sized enterprises (SMEs) has been well documented (see, e.g., Black and Gilson, 1998; Gompers and Lerner, 1999; Kortum and Lerner, 2000). VCs play a significant role in enhancing the value of their entrepreneurial investments (e.g., Sahlman, 1990; Sapienza *et al.*, 1996; Gompers and Lerner, 1999; Manigart *et al.*, 2002a,b; Leleux and Surlemount, 2003; Wright *et al.*, 2006). As VCs affect the product market strategies and outcomes of entrepreneurial companies, the variability of equity financing can also affect the performance of entrepreneurial ventures.

In response to the perceived importance of venture capital (“VC”) to the funding of entrepreneurial companies, many governments have mounted programs that seek to foster VC financing. Such programs have been the subject of previous scholarly examination; see, e.g., Cressy (2002), Keuschnigg and Nielsen (2003a), Kanninen and Keuschnigg (2004), Carpentier and Suret (2006), Bruton, Fried and Manigart (2006). Lerner (1999) evaluates the success of the US government’s SBIR program. Lerner (1999, 2002), Cressy (2002), Leleux and Surlemount (2003) and Cumming and MacIntosh (2006, 2007), among others, discuss the appropriate role of governments in private equity markets¹, and consistently argue that government programs ought to complement – and not compete with – wholly private VC investments. Recent advances in the literature have considered the role of informed versus uninformed VCs and government policy towards VC (Kanninen and Keuschnigg, 2003, 2004; Keuschnigg, 2004; Keuschnigg and Nielsen, 2001, 2003a,b), and come to similar conclusions in regards to the role of complementarities between the public and private sector in the support of VC. In short, as the social rate of return to VC is greater than the private rate of return, VC markets warrant public subsidies if such subsidies facilitate and do not crowd out private investment.

This paper examines one such government policy initiative – the Australian Pre-Seed Fund (PSF) program. The PSF program was announced in January 2001 as part of the Australian Government’s *Backing Australia’s Ability* package of funding initiatives. PSFs are derived from a mix of public and private investors. PSFs are established as private funds with direct subsidization from the Government of Australia. The Government is in effect a limited partner alongside other institutional investors. Terms provided tend to be more favourable than that which are available from institutional investors, and hence the competition among private VC fund managers to obtain a PSF licence is intense. In exchange for government support, PSFs must operate under special covenants

¹ For the purposes of this paper, the term private equity is used to encompass both the earlier stages of VC financing and also later stage private financing, including but not limited to mezzanine, buy-outs and turnaround financing.

such as the use of funds for investment in nascent enterprises (among other things discussed herein). The private VC fund managers that operate a PSF may nevertheless continue to operate their other VC funds which do not receive any government subsidization. The Australian Government has committed capital of \$72.7 million to the four licensed PSFs which, when combined with capital of \$31.4 million from private sector investors, universities and public sector research agencies, provides a total pool of \$104.1 million under the PSF program to assist the commercialisation of eligible research and development activities.

The fundamental issue examined in this paper is whether the structure of the PSF program facilitates an important and valuable source of value-added capital to nascent high-tech entrepreneurial companies. We also assess whether similar government programs for entrepreneurial finance can operate in harmony or whether they compete with each other. In the Australian context, the PSF program was established in 2002, but a similar program known as the Innovation Investment Fund (IIF) Program had been previously established in 1997.

In brief, the Venture Economics data indicate PSFs are the primary provider of seed stage VC in Australia, but are not more likely to invest in high-tech companies than other types of private or public VC funds. PSFs have smaller portfolios (number of investees) per manager than other types of private and public VC funds, and are more likely to invest in companies that are resident in the same state, but do not stage and syndicate more frequently than other types of private and public VC funds. Overall, therefore, the structure of the program has given rise to mixed performance in terms of governance provided to nascent entrepreneurial companies.

This paper further presents evidence consistent with the view that the PSF program diminishes the incentives for IIFs to invest in seed ventures, and hence competing government initiatives appear to be crowding out one another. It is further noteworthy that other confidential data provided by the Government of Australia is suggestive that one PSF has outperformed the other funds in regards to the investee company patents and financial statement performance, while at the same time having invested less money and charging lower management fees. Hence, an important lesson from the PSF program is that the impact of government sponsored VC funds depends not only on the design of the program but also on the selection of the VC managers carrying out the investments.

This paper is organized as follows. Section 2 provides a background and context of the Australian VC programs relative to that in other select Commonwealth countries. Section 3 discusses the PSF program details. Section 4 explains of the evaluation criteria for the PSFs and introduces the data used to evaluate the program. Statistical and econometric analyses are provided in section 5.

Section 6 discusses related information about PSF investees as provided by the Government of Australia, and future research. Concluding remarks follow in the last section.

2. Background and Context

The Pre-Seed Fund (PSF) program is one of five primary VC programs in Australia. As described in section 3 below, the focus of the PSF program is on the earliest stage VC investments in new seed stage companies. The PSF program was initiated under a competitive selection process whereby private VC fund managers compete for a license to operate the program. Awards were granted to four fund managers in 2002.

The other VC programs in Australia comprise the Innovation Investment Fund (IIF) program, the Renewable Energy Equity Fund (REEF) program, the Pooled Development Fund (PDF) program (1992), and the Venture Capital Limited Partnership program (2002).

The IIF program is similar in structure to the PSF program in that it is based on a competitive selection process for licenses to operate funds partly provided by the Government. While the focus of PSFs is on brand new seed stage companies, IIFs also invest in early and expansion stage companies. The first round of the program was announced in the Government's *Small Business Statement* in March 1997 and provided \$130 million, which has been matched on the basis of a Government to private sector capital ratio of up to 2:1. Five licensed funds were established in round one (A&B, AMWIN, Momentum, GBS (formerly Rothschild) and Coates Myer) and became operational during 1998. The second round of the IIF program was announced in the Government's industry statement, *Investing for Growth*, in December 1997 and funding of \$90.7 million was provided, also matched by private sector capital on the basis of a Government to private sector capital ratio of up to 2:1. The Government to private sector capital ratio was a competitive element in the selection of the potential round two funds. Four funds were licensed under round two (Foundation, Nanyang, Neo (formerly Newport) and Start-up) and became operational in 2001. The nine licensed funds have total capital of \$385.05 million, of which the Australian Government is contributing \$220.7 million and the private sector \$137.35 million.

The Renewable Energy Equity Fund (REEF) program is similarly structured with a competitive selection process for private VC fund managers to compete for the supply of funds from the Government to subsidize a new fund. Funding for the specialist renewable energy VC fund, subsequently known as REEF, was announced in November 1997 in the Australian Government's statement, *Safeguarding the Future: Australia's Response to Climate Change*. After a competitive selection process held in 1999, the Government awarded a single ten-year licence to the specialised

private VC fund manager, CVC REEF Limited, which became operational during 2000. The Australian Government has committed capital of \$17.7 million to REEF, which has been matched by \$8.9 million from private sector investors on the basis of a Government to private capital ratio of 2:1 to provide a total capital pool of \$26.6.

The Venture Capital Limited Partnerships (VCLP) program was established in 2002. It enables funds to be structured as limited partnerships, consistent with international standards. A major intended benefit of the program is the attraction of capital from international institutional investors who are more familiar with the limited partnership structure than any other.

Black and Gilson (1998) argue that the strength of a country's stock market is vital for a healthy VC market, and this idea is empirically supported by the work of Jeng and Wells (2000). In a more recent cross-country study, Leleux and Surlemont (2003) and Armour and Cumming (2006) provide empirical evidence industry-wide data regarding the performance of public policy initiatives towards VC across Europe and North America. This evidence generally indicates temperate bankruptcy laws stimulate entrepreneurial demand for VC; that government programs more often hinder than help the development of private equity markets; and that the legal environment (including tax incentives) matters as much as the strength of stock markets in stimulating the demand for, and supply of, VC. The importance of low capital gains taxes has also been stressed by Poterba (1989a,b), Keuschnigg (2004), Keuschnigg and Nielsen (2001, 2003a,b), and Parker (2007).

The structure of government policy programs to VC finance in Canada, Germany, Israel, the UK and the US has been described in OECD (1996), Cressy (2002), Lerner (1999, 2002), and Gilson (2003). In brief, prior research has established that government programs in the US and Israel that partners government funds with private fund managers has been rather successful. By contrast, government VC programs that compete with private VC funds have resulted in less successful outcomes in Canada (Cumming and MacIntosh, 2006, 2007) and Europe (Leleux and Surlemont, 2003; Amour and Cumming, 2006; Tykvova', 2000; Tykvova' and Walz, 2006).

In the Australian context, Cumming (2007) provides an empirical analysis of the IIF program and shows that the IIF program has (based on data to 2005(Q1)) significantly contributed to the Australian VC industry. The design of the program with a mix of private and Government support has successfully given rise to investments that jump-started the Australian VC industry in 1997, consistent with the design of the US and Israeli VC programs.

This paper extends the analysis of government sponsored VC programs by analyzing the Australian PSF program which is focused on more nascent stage entrepreneurial ventures than the IIF

program. Among other things, we provide generalizable insights into whether the PSF program design is appropriate for nascent stage entrepreneurial ventures, whether similar government programs for entrepreneurial finance operate in harmony or whether they compete with each other, and whether the success of the program depends on other factors other than the program design, such as the fund managers selected to carry out the investments. The institutional details of the PSF program are discussed in the next section. Empirical analyses thereafter follow in the next sections.

3. The Institutional Details of the Australian PSF program

3.1. Establishment

The PSF program was announced in January 2001 as part of the Australian Government's *Backing Australia's Ability* package of funding initiatives. After a competitive selection process held in 2001/2002, the Government established four licensed funds (A&B, GBS, Sciventures and Starfish), which became operational towards the end of 2002. The Government has committed capital of \$72.7 million to the four licensed funds which, and combined with capital of \$31.4 million from private sector investors, universities and public sector research agencies, provides a total pool of \$104.1 million under the PSF program to assist the commercialisation of eligible R&D activities. One of the funds (GBS) specialises in life sciences, one (A&B) has a focus on information and communications technologies, and the remaining two (Sciventures and Starfish) invest in a broad range of technologies.

3.2. Objectives

The objectives of the PSF program are:

- to assist the commercialisation of R&D activities undertaken by universities and public sector research agencies by providing financial and managerial advice;
- to encourage private sector investment in R&D activities undertaken in universities and public sector research agencies for commercialisation;
- to build linkages between universities, public sector research agencies, the finance community and business for the commercialisation of R&D activities;
- to build entrepreneurial and intellectual property skills in Australian universities and public sector research agencies; and
- to encourage researchers in universities and public sector research agencies to consider the commercial opportunities of their research discoveries.

3.3. Structural Features

The delivery model for the PSF program is similar to that of the IIF program described in section 2 above. In view of the higher risk of seed stage investments made under the PSF program, however, the Australian Government is providing up to three quarters of the total capital and has agreed to a more favourable profit distribution for private investors and a higher rate of management fees than under the IIF program. Annual fees for the four PSF managers range from 3% to 3.5% of committed capital (compared to 2.5-3% for IIFs). Government to private sector capital ratios and management fee levels were competitive elements in the PSF selection process.

The total pool of committed capital available under the PSF program (\$104.1 million) will be distributed among four funds, ranging in size from \$20 million to \$33 million, to be drawn down for making investments and paying management fees. The funds are administered by licensed private fund managers who make all investment decisions, subject to the terms of their licence agreements with the Australian Government and other governing documents.

Key elements of the PSF's operating requirements are:

- the ratio of Government to privately sourced capital must not exceed 3:1;
- investments will generally be in the form of equity and must only be in eligible companies or projects spinning out of universities or public sector research agencies;
- at least 60% of each PSFs' committed capital must be invested within five years;
- unless specifically approved by the Industry Research and Development Board, a fund must not invest more than \$1 million in any eligible company or eligible project;
- distribution arrangements provide for:
 - an amount equal to each PSF's total committed capital to be returned to the Government and other investors in accordance with the amounts they have subscribed; and
 - any further amounts to be distributed between the fund manager and the other investors, *excluding the Australian Government*, in an agreed manner as set out in the fund's governing documents; and
- the funds established under the PSF program will have a term of ten years, after which they will be closed in a commercially prudent manner.

To be eligible for investment under the PSF program, *companies* must be commercialising research and either:

- be controlled by a university, public sector research agency or qualifying researcher; or
- using intellectual property that is at least 50% owned by a university, public sector research agency or qualifying researcher.

Eligible companies must also be incorporated in (and operate substantially in) Australia and must not have generated any sales revenue.

Eligible *projects* must:

- have at least 50% of their intellectual property owned by a university, public sector research agency or qualifying researcher; and
- be controlled or supervised by a university or public sector research agency.

Eligible projects must also be undertaken in Australia and must not have generated any sales revenue.

3.4. Intended Benefits

Under the PSF program, it is intended that the funds will acquire an equity interest in the eligible companies and projects they are supporting. They will ultimately divest their interest in successful companies and projects to later stage investors and will seek, after distribution of any proceeds, to make a profit through capital gain. In return for any eventual proceeds from their investment in eligible companies and projects, the fund managers will provide management and technical advice to develop the commercial potential of the technology being supported. The next sections of this paper evaluate the extent to which the program has achieved its intended benefits as at 2005 (Q4).

4. Data and Summary Statistics

The data used for the evaluation of the PSF program are publicly available data from Venture Economics. The publicly available data from Venture Economics provide information on 970 entrepreneurial companies spanning 1982(Q1) to 2005(Q4). The data are used to provide rigorous statistical and econometric analyses of the Australian VC market, and the impact of the PSF program. For these data, we are able to ascertain the performance of PSFs collectively along several dimensions, including:

- The propensity of PSFs to take on risk by investing in seed stage and high-tech investments;
- The propensity of PSFs to screen, monitor and add value to investee companies through staging, syndication, and portfolio size per fund manager;
- The regional distribution of investments and probability that the investor and investee are based in the same state.

We begin describing the data with the aid of a number of graphs. We begin with a time-series analysis of seed investments (Figure 1) and high-tech seed stage investments (Figure 2; and see Figure 3 for specific industry breakdowns). The time series data from 1982(Q1)-2005(Q4) in Figure 1 indicate a comparative dearth of financings in seed venture investments prior to the introduction of the IIF program, which became operational in 1998. The PSF program became operational in late 2002.² The time series data indicate an increase in seed stage PSF investments in 2003, but a reduction in seed stage IIF investments in 2003. This is consistent with prior work which shows that IIFs started in 2001/02 are less likely than IIFs started in 1997/98 to invest in seed stage companies (Cumming, 2007). This evidence is highly consistent with the interpretation that PSFs compete with (and possibly crowd out) IIF investments at the seed stage. As well, these results cannot be explained by a winding down of IIF investments, as 5 IIFs were introduced in 1998 and another 4 IIFs were introduced in 2002, and IIFs also have a mandate to invest at the seed stage. An important policy lesson to take away from these data is the need for a complementary design of government VC programs, and not similar programs with competing objectives.

[INSERT FIGURES 1 – 3 ABOUT HERE]

The additional details in the Venture Economics data are provided in the tables. The available variables in the data are defined in Table 1. Summary statistics of the data are provided in Tables 2-4. The correlation matrix in Table 3 and the comparison of means and medians tests in Table 4 indicate PSFs, and funds affiliated to organizations that operate a PSF, are more likely to finance seed and early stage companies than other types of investors, and these differences are significant at the 5% level of significance.

[INSERT TABLES 1 – 4 ABOUT HERE]

The univariate comparison tests in Table 4 indicate the median PSF investee company operates in an industry with higher market/book ratios (generally riskier high-tech industries), although the average PSF investee company operates in a lower market/book industry. Funds affiliated to PSFs are more likely to finance companies that operate in industries with higher market / book ratios than other types of investors (both in terms of means and medians), and these differences are significant at the 5% level of significance.

² Note that while the PSFs first started investing in 2002, Figures 1 and 2 indicate PSF investments in 2000 and 2001. The reason for this discrepancy is that Figures 1 and 2 are graphed for the date of first VC investment in the particular portfolio company. Hence, the earlier VC investments are investments already in the pipeline before the PSF was “officially” announced as a beneficiary of the program or received its funding from the Australian Government.

Table 4 indicates PSFs, and funds affiliated to organizations that operate a PSF, stage capital contributions (thereby monitor and add value to investees) more than other types of investors, and these differences are significant at at least the 5% level of significance. The univariate comparison tests also indicate PSFs, and funds affiliated to organizations that operate a PSF, syndicate investments (which is suggestive of better screening and more value added to investees) than other types of investors, and these differences are significant at the 5% level of significance.

The univariate comparison tests in Table 4 indicate no statistically significant difference in the exit outcomes (IPOs, acquisitions, write-offs) for PSFs, funds affiliated with PSFs and other types of private and public VC funds. The Venture Economics data indicate PSFs have had 0 IPOs, 1 acquisition exit and 0 write-offs. The Australian Government's Department of Industry, Tourism and Resources data indicate 4 write-offs are likely to be forthcoming in the near future among the PSF investments. There have not been a sufficient number of PSF exits to carry out multivariate empirical analyses of exit outcomes and/or share price analyses of IPO exits. These findings on exit outcomes are best viewed as preliminary, as many investments in the sample (particularly PSF investments) have yet to be exited. In the econometric analyses below, note that we do not distinguish between the 4 different PSFs as there are not enough observations per PSF to enable statistically significant comparisons. As such, we group all types of PSFs together and treat all equally in the multivariate analyses in section 5.

5. Econometric Regression Analyses

The econometric regression analyses in this section formally test the evaluation criteria that were enumerated above in section 4. The focus of the analysis is on the effect of the indicator variables for fund types (PSFs, funds in organizations that are affiliated with a PSF, and non-PSFs).³ Control variables are provided for other factors pertaining to differences over time (e.g., to capture trends in the degree of learning and skills over time, etc.), market conditions (as proxied by the Australian MSCI Index, and by indicator variables for investments and exits during the Internet bubble in 1999 and 2000), among other things discussed below. The tests of the different evaluation criteria proceed sequentially in the following subsections: 5.1 stage of development at first investment; 5.2 industry; 5.3 staging; 5.4 syndication; 5.5 portfolio size; 5.6 exit outcomes; and 5.7 geographic scope

³ Separate dummy variables are not used for the four different PSFs since the PSFs are compared on a relative basis to other fund types and not specific funds within other fund types, and hence it is appropriate to only consider dummy variables for the type of fund and not specific funds within one class of funds. The econometric evidence herein thereby shows the performance of the average PSF. As well, each individual PSF has not had a sufficient number of investments that give rise statistically significant interpretations in the regression analyses for separate dummy variables (but collectively for all four PSFs there are a sufficient number of PSF investments).

of investments. The econometric methods and robustness checks are described within each subsection. A discussion of companion evidence from other sources of information is provided in section 6.

5.1. Stage of Development

Table 5 provides logit regressions of the probability of investment at different stages of entrepreneurial company development. The left-hand-side variables take on the value one where the company was financed at the indicated stage of development (seed stage in Model (1) in Table 5, early stage in Models (2), etc.) The logit regressions are a widely accepted methodology for dependent variables that are binary (taking on the value of either 0 or 1) (see, e.g., Judge *et al.*, 1988). Alternative specifications were excluded for reasons of conciseness, but are available upon request (the reported results are extremely robust).

[Table 5 About Here]

Positive and statistically significant coefficients indicate that fund type is more likely to finance an entrepreneurial company at a particular stage of development. Note as well that the numbers indicated directly reflect the economic significance of the results; for example, the numbers indicate the probability that a fund type will finance an entrepreneurial company at the particular stage of development for each of the models indicated. Moreover, note that the fund type dummy variable excluded from the regressions is the category of “private” VC funds (i.e., all of the unit trust funds and other private funds that are not corporate funds, investment bank affiliated funds, etc.; see Table 1 for definitions). Hence, the numbers in Table 5 next to the fund variables reflect the likelihood of an investment at a different stage in comparison to the base case of the class of other private funds.

The logit regressions in Table 5 indicate (at at least the 5% level of significance) PSFs are 16% more likely to finance seed stage companies than IIFs, and 61% more likely to finance seed stage companies than other types of private VC funds (Model 1). PSFs are 30% less likely than IIFs to finance companies in the early stage of development, and neither more nor less likely to finance early stage companies than other types of private VC funds (Model 2). PSFs are 29% less likely to finance expansion stage companies than other types of private VC funds (Model 3). PSFs do not finance late stage companies or buyouts (Models 4 and 5). Private VC funds that are affiliated to organizations with a PSF are neither more nor less likely to finance seed, early or expansion stage investments than other types of private VC funds (Models 1-3). Overall, these findings indicate PSFs have significantly contributed to the financing of seed stage companies. However, the findings also indicate that managers that operate PSFs are not independently investing in seed stage companies with their other funds (non seed stage centric) that do not benefit from the capital from the Government.

These findings are robust in a multivariate econometric setting which controls for other types of private VC funds, market conditions and trends in financing patterns over time. Generally, the data indicate universities and corporate funds are more likely to finance earlier stage companies and less likely to finance late stage companies and buyouts.⁴ Investment banks and other financial institutions are more likely to finance buyouts. Over time, investments in seed stage companies and buyouts have become more common. During the bubble period investments in buyouts were less common, and late stage investments were more common (this suggests investors benefited from easier exit conditions during the bubble by investing in late stage companies that were close to being able to go public). Buyout investments have been more common in years in which the MSCI index has been higher.⁵

5.2. Industry

Table 6 presents logit regression results for industry choices among the VC and private equity investors in Australia. The methodology used is very similar to that used in Table 5, as described above in subsection 5.1. The left-hand-side dependent variable is a dummy variable for the different industries indicated for Models (6) – (9) in Table 6. The right-hand-side explanatory variables in Table 6 are the same as those reported in Table 5 and discussed above in subsection 5.1. In addition to the logit regressions, Table 6 also presents an OLS regression for the determinants of the industry market/book variable (Model 10). Riskier and high-tech industries have higher market/book ratios (Fama and French, 1993, 1995; Gompers and Lerner, 1999). The industry market/book regressions therefore facilitate a useful robustness check as to the degree to which different types of VC funds in Australia undertake different risks.

[Insert Table 6 Here]

The multivariate empirical analyses in Table 6 indicate, relative to other types of private VC funds, PSFs are 19% more likely to finance biotech/medical investees than other types of private VC funds, although this difference is only significant at the 10% level of significance. PSFs, and funds affiliated to PSFs, are neither more nor less likely to finance companies that operate in high-tech industries than private VC funds.

⁴ Note that the dummy variable for universities are excluded for the late and buyout stage regressions (Models (4) – (5)) because there were not university investments at those stages; hence the variable had to be excluded to avoid perfect collinearity. This collinearity issue likewise applies for the other variables excluded in each of the regression tables.

⁵ This result is not attributable to correlation between the MSCI index and the bubble investment year dummy variable. The results are very robust to the inclusion/exclusion of either of these variables.

Overall, these findings indicate PSFs have not significantly contributed to the financing of high-tech companies relative to other types of private and public VC funds. Overall, the findings also indicate that managers that operate PSFs are also not more likely to finance high-tech companies with their other companion private VC funds relative to other non-affiliated private VC funds. Nevertheless, as indicated in subsection 5.1, PSFs are more likely to finance seed stage companies.

Some of the control variables for other types of VC funds, patterns over time and market conditions are statistically significant. For example, investment banks and other financial institutions are more likely to invest in Internet companies, but less likely to invest in computer hardware industries. Corporate funds are more likely to invest in computer industries. IIFs are 6%, 14% and 13% more likely than PSFs to finance companies in the biotech/medical, computer and Internet industries. We may infer that the IIF program has been more successful than the PSF program (at least as at 2005 Q4) in stimulating high-tech investment in Australia.

5.3. *Staging*

Model (11) in Table 7 presents a Poisson regression of staging frequency. The Poisson regression model is used to appropriately account for the distribution of the dependent variable: many investees in the data only have one or two staged financing rounds, and fewer investees are staged more frequently (Table 2 indicates the maximum number of staged rounds for one investee was 15 rounds, while the median number of rounds was only 1 round). The explanatory variables in Table 7 are very similar to those used in Tables 5 and 6 as described above. In addition, explanatory variables are also included for investee characteristics such as stage of development and capital requirements. These extra explanatory variables are appropriate to include, and have been included in seminal work on topic (Gompers, 1995; Gompers and Lerner, 1999), because less developed investee companies require more monitoring and value-added advice from their investors. Moreover, by including controls for stages of development alongside controls for types of investors, we ascertain in the regressions the marginal monitoring and effort provided by the different investee types while accounting for the need for monitoring and effort required by the investee company.

[Table 7 About Here]

Table 7 Model (11) indicates that, accounting for other variables, there are no pronounced differences in the staging frequency between PSFs, funds affiliated with IIFs, and other types of private VC funds. IIFs, and funds affiliated with IIFs, by contrast, are more likely to stage capital commitments than PSFs and funds affiliated with PSFs. Overall, these findings on staging indicate IIFs and funds affiliated with an IIF monitor more intensively, thereby adding value to their investee

companies relative to other types of private VC funds, but the same cannot be said of the PSFs. While the univariate tests described in section 4 were supportive of higher staging frequency among PSF and funds affiliated to PSFs, the multivariate analyses indicate PSFs and funds affiliated with PSFs do not monitor investees more intensively relative to other types of private and public VC funds. By contrast, IIFs stage more frequently than PSFs, and those differences are robust to the multivariate controls.

5.4. Syndication

Table 7 Model (12) analyses syndication frequency for each of the investee companies in the data. The Poisson regression methodology is also used for syndication frequency for the same reasons that it was used for staging frequency (few syndicated investors are observed very often, and many syndicated investors are rarely observed; see Table 2). The same explanatory variables are used for syndication as in staging. Table 7 Model (12) indicates (at at least the 5% level of significance) PSFs and funds in organizations affiliated with PSFs on average are not more likely to have syndicated investors than other types of funds. IIFs and funds affiliated with IIFs, by contrast, are more likely to syndicate than PSFs and other types of private and public VC funds. IIFs are more likely to have an average of one extra syndicated investor for each investee relative to PSFs and other types of private and public VC funds.

It has been well established that syndication enhances investment returns through better screening and more value-added advice provided to investees (Lerner, 1994; Lockett and Wright, 1999, 2001; Wright and Lockett, 2003). Overall, these findings on syndication are suggestive that PSFs and funds affiliated with a PSF do not screen better potential investees and add more value to their investee companies relative to other types of private funds. By contrast, IIFs do syndicate more frequently than PSFs, and those differences are robust to controls in a multivariate setting.

5.5. Portfolio Size (Number of Investee Companies) / Fund Manager

Table 7 Model (13) further examines the factors that lead to differences in the number of companies financed by a fund. The dependent variable is the number of companies financed divided by the number of fund managers. Unlike all of the other regressions where each observation is a different entrepreneurial company, each observation in Table 7 is a different fund; hence the number of observations is 164 for Model (13) in Table 7.⁶

⁶ In total, 280 different funds exist in the AVCAL data. Only 164 funds are used in Models (5) and (6) in Table 5 because the data do not comprise a complete set of details for 116 of the 280 funds.

The portfolio size regressions in Table 7 Model (13) use a Box-Cox transformation of the right-hand-side variables with a heteroscedastic specification. Right-hand-side variables include all the entrepreneurial company and transaction characteristics discussed above. Each variable is defined as the proportion of the total VC fund portfolio with the particular characteristic. For example, the variable “Seed” is the proportion of all investments in a VC fund’s portfolio that are at the seed stage. Dummy variables are included for the type of VC fund. As well, a variable that measures the years and months of the investment duration (time from first investment to date of fund closing, or 2005(Q4) not closed) is used (for longer horizons we expect larger portfolios as a fund becomes fully invested). Transformations to independent variables (other than the dummy variables) are presented to capture nonlinearities in the relationship between portfolio size and the explanatory variables. The intuition underlying the possible existence of nonlinearities is explained in Cumming (2006).

Following the conventional notation, λ is used to denote the transformation variable for the right-hand-side variables. The transformation yields a convex relation when $\lambda > 1$, and a concave relation when $\lambda < 1$. Maximum likelihood optimization and grid searches were used to optimize the value of the loglikelihood function to ascertain the appropriate value for λ given the data. The Box-Cox grid search was sufficiently broad (-2 to +2) to allow for the possibility of concavity (diminishing returns to scale in the number of investees) as well as convexity (increasing returns to scale in the number of investees). The Box-Cox transformation is as follows:

$$B(x, \lambda) = \begin{cases} \frac{x^\lambda - 1}{\lambda} & \text{when } \lambda \neq 0; \\ \log(x) & \text{when } \lambda = 0. \end{cases}$$

where x represents the independent variables, and $\log(x)$ is the limit of $(x^\lambda - 1)/\lambda$ as $\lambda \rightarrow 0$. Note that the transformation is applied to variables that are *not* dummy variables.

The heteroscedastic specification is as follows: $\text{Var}(\varepsilon) = (\sigma^2)[w^2]^{(\lambda)}$, where w =fund duration. This heteroscedastic specification was employed to account for the possibility of unobserved observations in the data (i.e., the fact that not all the funds are fully invested; see Cumming, 2006). The likelihood dominance criterion indicated this heteroscedastic Box-Cox transformation outperformed other specifications (not reported for conciseness).

The portfolio size regressions in Model (13) in Table 7 indicates (at at least the 5% level of significance) that PSFs finance on average 0.94 fewer companies per manager than other types of private VC funds, and 0.58 fewer investees per manager relative to IIFs. Funds in organizations affiliated with a PSF do not have significant differences in portfolio size per manager relative to private VC funds. Overall, these findings on portfolio size per manager are suggestive that PSFs spend more time screening potential investees companies relative to other types of private and public

VC funds, and relative to IIFs (consistent with the theoretical work of Kanninen and Keuschnigg, 2003, 2004, and Keuschnigg, 2004). However, taken in conjunction with the staging and syndication evidence, we cannot conclusively interpret this evidence as suggestive of greater value-added among PSF investors relative to other types of private and public VC funds.

5.6. IPO, Acquisition and Write-off Exit Outcomes

The vast majority of the PSF investments have not yet been exited. The Venture Economics data indicate PSFs have as yet had 0 IPOs, 1 acquisition exit and 0 write-offs. The Australian Department of Industry, Tourism and Resources has indicated that 4 write-offs are likely to be forthcoming in the near future among the PSF investments. There have not been a sufficient number of PSF exits to carry out multivariate empirical analyses of exit outcomes and/or share price analyses of IPO exits; nevertheless, some summary statistics were indicated above in Tables 2-4.

5.7. Geographic Scope of Investments

Figure 4 presents a graphical analysis of the regional distribution of PSF investees. The data indicate 49% of PSF investees are located in Victoria, 22% in New South Wales, 11% in Queensland, 11% in the Capital Territory, and 7% of the investees' location is unknown. Table 7 Model (14) provides companion regression evidence on the likelihood that the investee and investor are resident in the same state. The multivariate empirical analyses in Model (14) indicates PSFs are 24% less likely to invest in a company that is not based in the same state relative to private VC funds and IIFs. This finding is expected since seed stage investments need to be regionally proximate to enable the investor to screen and incubate / add value to the investee.

[INSERT FIGURE 4 ABOUT HERE]

In sum, the data indicate PSFs are the primary provider of seed stage VC in Australia, but PSFs are not more likely to invest in high-tech companies than other types of VC funds. PSFs have smaller portfolios (number of investees) per manager than other types of VC funds, and are more likely to invest in companies resident in the same state, but do not stage and syndicate more frequently than other types of VC funds. The next section discusses companion evidence on PSF investments.

6. Related Information and Future Research

The publicly available data from Venture Economics in this study enabled a comparison of different VC-backed companies depending on the type of investor along several dimensions: stage of

development, industry focus, staging, syndication, regional analyses, etc., as analyzed above. Systematic data collection on patents, financial statement figures of investees, and fees, capital commitments and drawdowns of fund managers at all types of different VC funds would provide additional insight for assessing the economic impact of government sponsored VC funds.

It is important to note that private confidential data (that cannot be presented in this paper) from the Department of Industry, Tourism and Resources of the Government of Australia indicate that one PSF has received the least capital commitments and drawn down the least amount of capital relative to the other three PSFs. This particular fund also has lower management fees relative to the other funds. Further, this fund's investees have been quicker to file for patents and are the only investees to have obtained patents subsequent to PSF investment. The investees of this fund also have significantly higher R&D expenses than the investees of other PSFs. Overall, therefore, the private confidential data from the Department of Industry, Tourism and Resources is suggestive that one PSF is outperforming the other PSFs. These performance indicators are highly suggestive of likely future success potential (Gompers and Lerner, 1999). For example, prior research based on US data indicates VC-backed companies are more likely to have successful outcomes the greater the number of patents that they hold (Mann and Sager, 2006). Prior work is also consistent with the view that fund manager characteristics are important for the success of VC-backed investments (see, e.g., Hsu, 2004; Manigart *et al.*, 2007; Shepherd *et al.*, 2003, 2005; Zacharakis and Shepherd, 2001, 2003). As well, prior work has shows different types of programs may be more appropriate for different types of individuals (Westhead *et al.*, 2005).

As well, despite these measurable performance attributes that are superior among one of the PSFs, the unexited valuations of this PSF's investees is equal to that among two of the other PSF investees, and significantly lower than the one of the other PSF investees. Taken together with the other information on performance, this suggests that the unexited valuations among some of the other PSFs may in fact be inflated relative to their true valuation.

The confidential data provided by the Department of Industry, Tourism and Resources summarized in this section is available for the PSFs only. To empirically compare PSFs with other types of VC-backed investees in Australia, it is necessary to turn to the publicly available data from Venture Economics. Those data enabled consideration of staging and syndication, among other things analysed in this paper, which provide indicative measures of performance (albeit perhaps more coarse measures relative to the confidential data described above). Further research on government VC programs could be fruitful with more detailed datasets that enable other comparisons of detailed information from both government (public) and non-government (wholly private) VC funds.

7. Summary and Conclusions

This paper studied the effectiveness of the PSF program in Australia for facilitating value-added entrepreneurial finance to nascent entrepreneurs. PSFs are established as private funds with direct subsidization from the Government of Australia, and run by private VC fund managers in cooperation with other industry partners. The participation of the Government as a source of capital for the PSF carries with it special terms and conditions intended to aid in the development of newly formed entrepreneurial companies. In addition to analyzing whether the PSF program design is appropriate for nascent stage entrepreneurial ventures, this paper also considered whether similar government programs for entrepreneurial finance operate in harmony or whether they compete with each other, and whether the success of the program depends on other factors other than the program design, such as the fund managers selected to carry out the investments.

The Venture Economics data analyzed in this paper indicate PSFs are more likely than other Australian Government programs such as IIFs to finance seed stage companies, but PSFs are less likely to finance high-tech companies. PSFs have smaller portfolios per fund manager than IIFs, but stage and syndicate less frequently than IIFs. Overall, therefore, the evidence is mixed in regards to the value-added provided by PSFs relative to IIFs. There is also suggestive evidence that the PSF program diminishes the incentives for IIFs to invest in seed stage ventures and hence different government programs appear to compete for deal flow.

As at 2005 Q4, confidential data provided by the Australian Government suggests that one PSF has outperformed the other 3 PSFs in regards to the investee company patents and financial statement performance, even though this PSF has invested less money and charged lower management fees than its 3 counterparts. Hence, an important lesson from the PSF program is that the impact of government sponsored VC funds depends not only on the design of the program but also on the selection of the VC managers carrying out the investments.

References

- Armour, J., and D.J. Cumming, 2006. The legislative road to Silicon Valley, *Oxford Economic Papers* 58, 596-635.
- Black, B. S., and Gilson, R.J. 1998. Venture capital and the structure of capital markets: banks versus stock markets. *Journal of Financial Economics* 47, 243-277.
- Bruton, G.D., V.H. Fried and S. Manigart, 2006. Institutional influences on the worldwide expansion of venture capital, *Entrepreneurship Theory and Practice* 30, 687-708.

- Carpentier, C., and J.-M. Suret, 2006. Bypassing the financial growth cycle: evidence from Capital Pool Companies, *Journal of Business Venturing* 21, 45-73.
- Cressy, R., 2002. Funding gaps: a symposium, *Economic Journal* 112, F1-F16.
- Cumming, D.J., 2006. The determinants of venture capital portfolio size: empirical evidence, *Journal of Business* 79, 1083-1126.
- Cumming, D.J., 2007. Government policy towards entrepreneurial finance: innovation investment funds. *Journal of Business Venturing* 22, 193-235.
- Cumming, D.J. and J.G. MacIntosh, 2006. Crowding out private equity: Canadian evidence, *Journal of Business Venturing* 21, 569-609.
- Cumming, D.J., and J.G. MacIntosh, 2007. Mutual funds that invest in private equity? An analysis of labour sponsored investment funds, *Cambridge Journal of Economics* 31, 445-487.
- Department of Industry, Tourism and Resources, 2004. AusIndustry Venture Capital Programs, September 21, 2004 document.
- Fama, E.F., and K.R. French, 1993. Common risk factors in the returns on stocks and bonds, *Journal of Financial Economics* 33, 3-56.
- Fama, E.F., and K.R. French, 1995. Size and book-to-market factors in earnings and returns, *Journal of Finance* 50, 131-156.
- Gilson, R.J., 2003. Engineering a venture capital market: lessons from the American experience, *Stanford Law Review* 55, 1067-1103.
- Gompers, P.A., 1995. Optimal investment, monitoring, and the staging of venture capital. *Journal of Finance*, 50, 1461-1489.
- Gompers, P.A. and J. Lerner, 1999. *The Venture Capital Cycle*. Cambridge: MIT Press.
- Hsu, D., 2004. What do entrepreneurs pay for venture capital affiliation? *Journal of Finance*, 59, 1805-1844.
- Jeng, L.A., and P.C. Wells, 2000. The determinants of venture capital funding: evidence across countries, *Journal of Corporate Finance*, 6, 241-289.
- Judge, G.G., W.E. Griffiths, R.C. Hill, H. Lutkepohl and T-C Lee, 1988. *Introduction to the Theory and Practice of Econometrics*, Second Edition, Wiley.
- Kanniainen, V., and C. Keuschnigg, 2003. The optimal portfolio of start-up firms in venture capital finance, *Journal of Corporate Finance* 9, 521-534.
- Kanniainen, V., and C. Keuschnigg, 2004. Start-up investment with scarce venture capital support, *Journal of Banking and Finance* 28, 1935-1959.
- Keuschnigg, C., 2002. Venture capital backed growth, *Journal of Economic Growth* 9, 239 – 261.
- Keuschnigg, C., 2003. Optimal public policy for venture capital backed innovation, CEPR Working Paper No. 3850.

- Keuschnigg, C., 2004. Taxation of a venture capitalist with a portfolio of firms, *Oxford Economic Papers* 56, 285-306.
- Keuschnigg, C., and S.B. Nielsen, 2001. Public policy for venture capital, *International Tax and Public Finance* 8, 557-572.
- Keuschnigg, C., and S.B. Nielsen, 2003. Tax policy, venture capital and entrepreneurship, *Journal of Public Economics*, 87, 175-203.
- Keuschnigg, C., and S.B. Nielsen, 2003b. Taxes and venture capital support, *Review of Finance* 7, 515 - 539.
- Keuschnigg, C., and Nielsen, S.B., 2004a. Progressive taxation, moral hazard, and entrepreneurship. *Journal of Public Economic Theory* 6, 471 – 490.
- Keuschnigg, C., and S.B. Nielsen, 2004b. Start-ups, venture capitalists and the capital gains tax, *Journal of Public Economics*, 88, 1011-1042.
- Kortum, S., and J., Lerner, 2000. Assessing the contribution of venture capital to innovation, *RAND Journal of Economics* 31, 647-692.
- Leleux, B., and B. Surlemont, 2003. Public versus private venture capital: seeding or crowding out? A Pan-European analysis. *Journal of Business Venturing* 18, 81-104.
- Lerner, J., 1994. The syndication of venture capital investments, *Financial Management* 23, 16-27.
- Lerner, J., 1999. The government as venture capitalist: The long-run effects of the SBIR program, *Journal of Business* 72, 285-318.
- Lerner, J., 2002. When bureaucrats meet entrepreneurs: the design of effective ‘public venture capital’ programmes, *Economic Journal* 112, F73-F84.
- Lockett, A., and M. Wright, 1999. The syndication of private equity: evidence from the U.K., *Venture Capital: International Journal of Entrepreneurial Finance* 1, 303 – 324.
- Lockett, A.; M. Wright, 2001. The syndication of venture capital investments, *Omega: The International Journal of Management Science* 29, 375-390.
- Manigart, S., A. Lockett, M. Meuleman, M. Wright, H. Landstrom, H. Bruining, P. Desbrieres, U. Hommel, 2002a. Why do European venture capital companies syndicate? Working Paper, Vlerick Leuven Gent Management School.
- Manigart, S., K. DeWaele, M. Wright, K. Robbie, P. Desbrieres, H.J. Sapienza, and A. Beekman, 2002b. The determinants of the required returns in venture capital investments: a five-country study. *Journal of Business Venturing* 17, 291-312.
- Manigart, S., V. Collewaert, M. Wright, S. Pruthi, A. Lockett, H. Bruining, U. Hommel, and H. Landstrom, 2007. Human capital and the internationalisation of venture capital firms, *International Entrepreneurship and Management Journal* 3, 109-125.
- Mann, R.J., and T.W. Sager, 2006. Patents, venture capital and software start-ups. Working Paper, University of Texas at Austin.
- Organization for Economic Corporation and Development, 1996. Government programmes for venture capital. Paris: Organization for Economic Corporation and Development.

- Parker, S., 2007. Law and the economics of entrepreneurship, *Comparative Labor Law and Policy Journal* 28, 695-715.
- Poterba, J., 1989a. Capital gains tax policy towards entrepreneurship, *National Tax Journal* 42, 375 – 389.
- Poterba, J., 1989b. 'Venture capital and capital gains taxation,' In: Lawrence H. Summers, ed., *Tax Policy and the Economy* 3, Cambridge: MIT Press, 47-67.
- Sahlman, W.A., 1990. The structure and governance of venture capital organizations, *Journal of Financial Economics* 27, 473-524.
- Sapienza, H., S. Manigart, and W. Vermeir, 1996. Venture capital governance and value-added in four countries. *Journal of Business Venturing* 11, 439-469.
- Shepherd, D.A., Armstrong, M.J., and M. Lévesque. 2005. Allocation of attention within venture capital firms: a queuing network model, *European Journal of Operational Research* 163, 545-564.
- Shepherd, D.A., A.L. Zacharakis and R.A. Baron, 2003. VCs' decision processes: evidence suggesting more experience may not always be better. *Journal of Business Venturing* 18, 381-401.
- Tykvova', T., 2000. Venture capital in Germany and its impact on innovation, Working Paper, Centre for European Economic Research (ZEW), University of Mannheim.
- Tykvova', T., and U. Walz, 2006. How important is participation of different VCs in German IPOs? *Global Finance Journal*, forthcoming.
- Westhead, P., D. Ucbasaran, and M. Wright, 2005. Policy toward novice, serial and portfolio entrepreneurs. *Environment and Planning C: Government and Policy* 22, 779-798.
- White, H., 1980. A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica* 48, 817-838.
- Wright, M., and A. Lockett, 2003. The structure and management of alliances: syndication in the venture capital industry. *Journal of Management Studies* 40, 2073-2102.
- Wright, M., B. Clarysse, A. Lockett, M. Binks, 2006. University spin-out companies and venture capital, *Research Policy* 35, 481-501.
- Zacharakis, A.L., and D.A. Shepherd, 2001. The nature of information and venture capitalists' overconfidence. *Journal of Business Venturing*, 16, 311-332.
- Zacharakis, A.L., and D.A. Shepherd, 2003. A non-additive decision-aid for venture capitalists' investment decisions. *European Journal of Operational Research* 162, 673-689.

Figure 1. Seed Stage Venture Capital Investments in Australia, 1982-2005

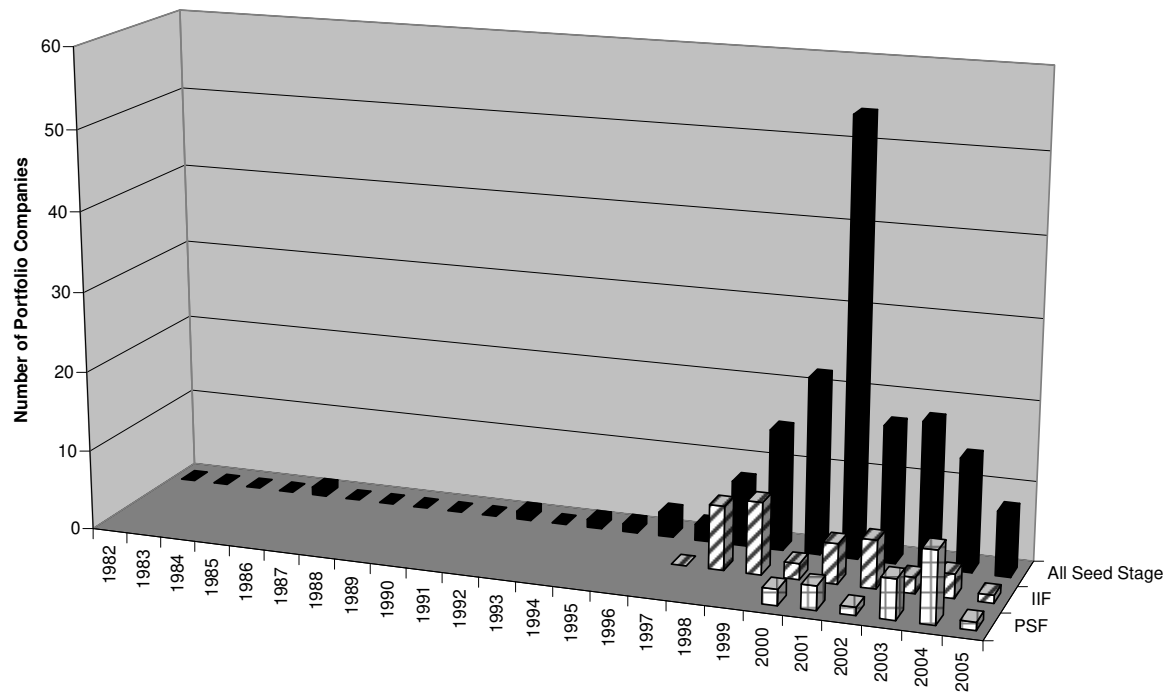


Figure 2. Seed Stage High-Tech Venture Capital Investments in Australia, 1982-2005

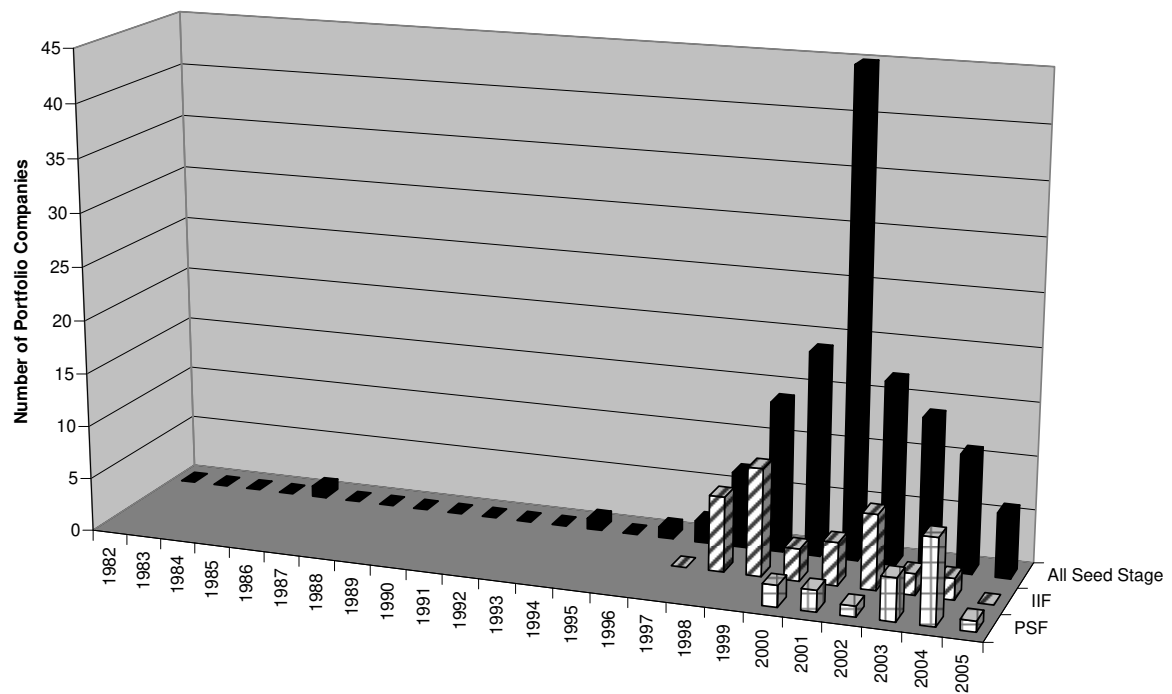


Figure 3. Industry Breakdown of PSF Investments as at 2005

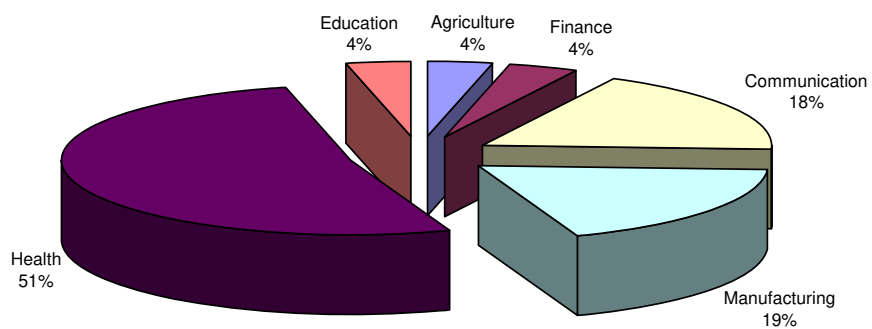


Figure 4. Regional Distribution of PSF Investments as at 2005

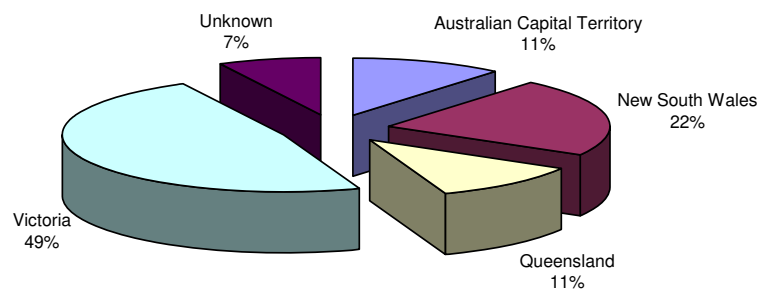


Table 1. Variable Definitions This table defines the variables used in the statistical and econometric analyses in the paper.	
Variable Name	Definition
PSF	A dummy variable equal to one for a PSF fund investment (and zero otherwise).
PSF Affiliate	A dummy variable equal to one for an affiliated PSF fund investment (and zero otherwise). An affiliated PSF fund is one which is part of a VC organization that has a companion fund that is a PSF fund, but the particular fund investing is not the companion PSF fund, but rather the companion fund.
IIF	A dummy variable equal to one for an IIF fund investment (and zero otherwise).
IIF Affiliate	A dummy variable equal to one for an affiliated IIF fund investment (and zero otherwise). An affiliated IIF fund is one which is part of a VC organization that has a companion fund that is an IIF fund, but the particular fund investing is not the companion IIF fund, but rather the companion fund.
Public VC	A dummy variable equal to one for a fund that receives at least in part government support (excluding IIFs) (and zero otherwise).
Government VC	A dummy variable equal to one a fund that is run by the government (excluding IIFs) (and zero otherwise).
Community Development VC	A dummy variable equal to one for a fund with a mandate of community development (and zero otherwise).
University	A dummy variable equal to one for a university fund investment (and zero otherwise).
Private	A dummy variable equal to one for a private fund investment (and zero otherwise).
Fund of Funds	A dummy variable equal to one for a fund of funds investment (and zero otherwise).
Green Investor	A dummy variable equal to one for a fund with a green (environmentally friendly) mandate (and zero otherwise).
Corporate VC	A dummy variable equal to one for a corporate fund investment (and zero otherwise).
Investment Bank	A dummy variable equal to one for an investment bank investment (and zero otherwise).
Financial Institution	A dummy variable equal to one for a financial institution investment (and zero otherwise).
Individual	A dummy variable equal to one for an individual investment (and zero otherwise).
Seed	A dummy variable equal to one for financing provided to companies carrying out research, and to assess and develop an initial concept (and zero otherwise).
Early	A dummy variable equal to one for financing provided to companies with product in testing and/or pilot production (and zero otherwise). The company may or may not be generating revenue, and has usually been in business less than 30 months.
Expansion	A dummy variable equal to one for financing provided to companies in need of development capital (and zero otherwise). The financing is provided for the growth and expansion of a company, which may or may not break even or trade profitably. Capital may be used to: finance increased production capacity; market or product development; provide additional working capital.
Late	A dummy variable equal to one if the company has reached profitable operating levels (and zero otherwise).
Buyout	A dummy variable equal to one for buyout financing either of MBO or MBI form (and zero otherwise). MBO: A buyout in which external managers take over the company. Financing is provided to enable a manager or group of managers from outside the target company to buy into the company with the support of private equity investors. MBI: A buyout in which the target's management team acquires an existing product line or business from the vendor with the support of private equity investors.
Turnaround	A dummy variable equal to 1 if the company is not earning its cost of capital.

Table 1. (Continued)	
Variable Name	Definition
Industry Market / Book	The industry market/book ratio for the company's primary industry.
Biotechnology / Medical	A dummy variable equal to one for an investee in the biotechnology and/or medical industries (and zero otherwise).
Computer	A dummy variable equal to one for an investee in the computer hardware and/or software industries (and zero otherwise).
Internet	A dummy variable equal to one for an investee in the Internet industry (and zero otherwise).
Communications	A dummy variable equal to one for an investee in the communications industry (and zero otherwise).
Semiconductor	A dummy variable equal to one for an investee in the semiconductor industry (and zero otherwise).
Domestic	A dummy variable equal to one for an Australian investee (and zero for a foreign investee).
Staging	The number of staged investment rounds in which VCs financed the company.
Syndication	The number of syndicated VC investors for all of the financing rounds.
Portfolio Size / Manager	The total number of investees financed by the fund (as at 2005(Q4)) divided by the number of fund managers.
Average \$ Size of Rounds	The average size of a financing round for the investee (in millions of 2005 \$Aus).
Average \$ Size of Firm	The average size of each investment in the firm (in millions of 2005 \$Aus).
Total \$ Amount Invested	The total amount invested in the firm (in millions of 2005 \$Aus).
Exit Status	A dummy variable equal to one if the investee has been exited (and zero otherwise).
IPO Exit	A dummy variable equal to one for an IPO exit (and zero otherwise).
Acquisition Exit	A dummy variable equal to one for an acquisition exit (and zero otherwise).
Leverage Buyout	A dummy variable equal to one for a leveraged buyout exit (and zero otherwise) in which the new company's capital structure incorporates a particularly high level of debt, much of which is normally secured against the company's assets.
Write-off	A dummy variable equal to one for a write-off exit (and zero otherwise).
MSCI Investment Year	The Morgan Stanley Capital International Index (MSCI) equity return (Australia) in the first round investment year.
MSCI Exit Year	The Morgan Stanley Capital International Index (MSCI) equity return (Australia) in the exit year.
MSCI Investment Horizon Annualized	The Morgan Stanley Capital International Index (MSCI) equity annualized return (Australia) over the investment horizon from the first round investment year to the exit year, or to 2005(Q4) if there has not been an exit.
MSCI Investment Horizon Gross	The Morgan Stanley Capital International Index (MSCI) equity gross return (Australia) over the investment horizon from the first round investment year to the exit year, or to 2005(Q4) if there has not been an exit.
Investment in Bubble Period	A dummy variable equal to one for a first round investment in 1999 or 2000 (and zero otherwise).
Exit in Bubble Period	A dummy variable equal to one for an exit in 1999 or 2000 (and zero otherwise).

Table 2. Summary Statistics

This table summarizes the data by the average, median, minimum and maximum values. The number of observations for each variable is indicated, as well as the number of observations where the dummy variables take on the value 1. The data span 1982 to 2005 (Q4). Source: Venture Economics.

	Mean	Number of Observations where Dummy Variable = 1	Median	Standard Deviation	Minimum	Maximum	Number of Observations
PSF	2.68E-02	26	0	1.62E-01	0	1	970
PSF Affiliate	4.64E-02	45	0	2.10E-01	0	1	970
IIF	6.39E-02	62	0	2.45E-01	0	1	970
IIF Affiliate	8.45E-02	82	0	2.78E-01	0	1	970
Public VC	9.79E-02	95	0	2.97E-01	0	1	970
Government VC	4.64E-02	45	0	2.10E-01	0	1	970
Community Development VC	1.03E-03	1	0	3.21E-02	0	1	970
University	1.86E-02	18	0	1.35E-01	0	1	970
Private	3.47E-01	337	0	4.76E-01	0	1	970
Fund of Funds	4.54E-02	44	0	2.08E-01	0	1	970
Green Investor	9.79E-02	95	0	2.97E-01	0	1	970
Corporate VC	2.37E-02	23	0	1.52E-01	0	1	970
Investment Bank	1.66E-01	161	0	3.72E-01	0	1	970
Financial Institution	2.09E-01	203	0	4.07E-01	0	1	970
Individual	2.58E-02	25	0	1.59E-01	0	1	970
Seed	1.70E-01	165	0	3.76E-01	0	1	970
Early	1.12E-01	109	0	3.16E-01	0	1	970
Expansion	4.55E-01	441	0	4.98E-01	0	1	970
Late	1.86E-02	18	0	1.35E-01	0	1	970
Buyout	1.41E-01	137	0	3.48E-01	0	1	970
Turnaround and Mezzanine	1.08E-01	105	0	3.11E-01	0	1	970
Industry Market / Book	4.59E+00	4457	3.955	1.96E+00	2.66	10.70	970
Biotechnology / Medical	1.37E-01	133	0	3.44E-01	0	1	970
Computer Hardware and Software	1.62E-01	157	0	3.69E-01	0	1	970
Internet	1.12E-01	109	0	3.16E-01	0	1	970
Communications	4.43E-02	43	0	2.06E-01	0	1	970
Semiconductor	3.51E-02	34	0	1.84E-01	0	1	970
Domestic	8.98E-01	871	1	3.03E-01	0	1	970
Staging	1.85E+00	NA	1	1.73	1	15	970
Syndication	1.63E+00	NA	1	1.12	1	9	970
Portfolio Size / Manager Average \$ Size of Rounds (millions)	6.55E+02	NA	2.47	4644.71	0.01	100282	852
Average \$ Size of Firm (millions)	5.44E+02	NA	2.65	3499.94	0.01	67511	852
Total \$ Amount Invested (millions)	7.76E+02	NA	4.115	5163.90	0.01	100282	852
Exit Status	2.49E-01	210	0	4.32E-01	0	1	845
IPO Exit	1.10E-01	107	0	3.13E-01	0	1	970
Acquisition Exit	3.55E-02	30	0	1.85E-01	0	1	845
Leveraged Buyout	6.27E-02	53	0	2.43E-01	0	1	845
Write-off	2.37E-02	20	0	1.52E-01	0	1	845
MSCI Investment Year	3.97E-02	NA	-0.01	1.64E-01	-0.27	0.50	961
MSCI Exit Year	1.45E-01	NA	0.15	1.94E-01	-0.13	0.45	208
MSCI Investment Horizon Annualized	9.08E-02	NA	0.08	6.52E-02	-0.12	0.36	961
MSCI Investment Horizon Gross	5.86E-01	NA	0.55	4.24E-01	-0.16	4.72	961
Investment in Bubble Period	6.49E-02	63	0	2.47E-01	0	1	970
Exit in Bubble Period	3.39E-01	329	0	4.74E-01	0	1	970
IPO Delist	2.18E-01	NA	0	4.17E-01	0	1	55
IPO Annualized Return	-3.16E-01	NA	-0.25	5.94E-01	-1	1.51	50
IPO Market Capitalization	2.61E+02	NA	48.2	1015.83	0	7337.50	53

Table 3. Correlation Matrix

This table presents correlations across selected variables presented in Tables 1 and 2. Correlations greater than 0.07 in absolute value are statistically significant at the 5% level of significance, and highlighted in underline font.

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)	(26)	(27)
(1)	PSF	1.00																										
(2)	PSF Affiliate	<u>0.14</u>	1.00																									
(3)	IIF	0.01	<u>0.27</u>	1.00																								
(4)	IIF Affiliate	<u>0.16</u>	<u>0.35</u>	<u>0.14</u>	1.00																							
(5)	Public VC	-	-	-	-	1.00																						
(6)	Government VC	-	-	-	-	-	1.00																					
(7)	Community Development VC	0.01	0.01	0.01	0.01	0.01	0.01	1.00																				
(8)	University	<u>0.27</u>	0.06	-	0.04	0.05	-	-	0.00	1.00																		
(9)	Private	<u>0.18</u>	<u>0.22</u>	<u>0.38</u>	<u>0.36</u>	<u>0.07</u>	<u>0.12</u>	0.03	0.01	1.00																		
(10)	Fund of Funds	-	0.00	-	-	-	-	-	-	-	1.00																	
(11)	Green Investor	0.06	0.01	<u>0.07</u>	<u>0.07</u>	0.01	<u>0.06</u>	0.01	0.05	0.01	<u>0.07</u>	1.00																
(12)	Corporate VC	-	-	-	-	-	-	-	-	-	-	-	1.00															
(13)	Investment Bank	<u>0.07</u>	0.01	0.06	<u>0.10</u>	<u>0.11</u>	<u>0.11</u>	0.02	0.06	<u>0.12</u>	0.05	0.01	0.00	1.00														
(14)	Financial Institution	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1.00												
(15)	Individual	0.03	0.01	0.04	<u>0.34</u>	0.04	0.03	0.01	0.02	0.09	0.03	0.05	0.02	<u>0.07</u>	0.05	1.00												
(16)	Seed	<u>0.29</u>	<u>0.15</u>	<u>0.30</u>	<u>0.11</u>	0.04	<u>0.28</u>	0.02	<u>0.25</u>	<u>0.18</u>	<u>0.11</u>	0.06	<u>0.12</u>	<u>0.14</u>	<u>0.14</u>	0.02	1.00											
(17)	Early	0.00	<u>0.09</u>	<u>0.19</u>	<u>0.10</u>	0.01	<u>0.10</u>	0.01	0.01	<u>0.19</u>	0.06	0.12	<u>0.09</u>	0.04	0.04	0.00	0.01	1.00										
(18)	Expansion	-	0.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1.00								
(19)	Industry Market / Book	-	<u>0.10</u>	<u>0.13</u>	<u>0.10</u>	<u>0.08</u>	<u>0.08</u>	<u>0.11</u>	0.01	0.06	<u>0.09</u>	0.03	0.06	<u>0.08</u>	0.02	0.03	<u>0.10</u>	<u>0.15</u>	0.05	1.00								
(20)	Domestic	0.06	0.06	<u>0.10</u>	<u>0.12</u>	<u>0.11</u>	<u>0.08</u>	0.01	0.05	<u>0.25</u>	<u>0.08</u>	<u>0.13</u>	0.05	<u>0.16</u>	<u>0.17</u>	0.05	<u>0.18</u>	<u>0.14</u>	<u>0.32</u>	<u>0.19</u>	1.00							
(21)	Staging	0.06	<u>0.27</u>	<u>0.34</u>	<u>0.31</u>	0.01	0.03	0.02	0.06	<u>0.33</u>	0.05	0.04	<u>0.12</u>	0.05	0.06	0.06	<u>0.12</u>	<u>0.25</u>	<u>0.20</u>	<u>0.11</u>	<u>0.13</u>	1.00						
(22)	Syndication	0.05	<u>0.19</u>	<u>0.30</u>	<u>0.18</u>	<u>0.12</u>	0.07	0.02	0.00	<u>0.29</u>	0.00	0.00	<u>0.10</u>	<u>0.19</u>	<u>0.10</u>	0.06	0.06	<u>0.16</u>	0.04	0.06	<u>0.09</u>	<u>0.48</u>	1.00					
(23)	Average \$ Size of Rounds	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
(24)	Exit Status	<u>0.08</u>	0.03	<u>0.11</u>	0.03	<u>0.09</u>	<u>0.13</u>	0.06	0.05	<u>0.07</u>	<u>0.12</u>	0.03	0.04	<u>0.16</u>	<u>0.16</u>	0.02	<u>0.16</u>	<u>0.07</u>	0.05	0.00	<u>0.19</u>	0.01	<u>0.12</u>	<u>0.07</u>	1.00			
(25)	IPO Exit	<u>0.07</u>	0.01	0.06	0.00	<u>0.14</u>	<u>0.08</u>	<u>0.10</u>	0.05	<u>0.09</u>	<u>0.07</u>	0.11	0.05	<u>0.10</u>	0.02	0.00	0.11	0.03	0.02	0.01	<u>0.14</u>	<u>0.07</u>	0.05	0.04	<u>0.65</u>	1.00		
(26)	Acquisition Exit	0.01	0.02	-	0.03	<u>0.07</u>	0.04	0.01	0.03	0.02	0.04	0.02	0.03	0.06	0.03	0.03	0.02	0.06	0.04	<u>0.08</u>	0.05	<u>0.08</u>	<u>0.09</u>	0.02	<u>0.33</u>	<u>0.07</u>	1.00	
(27)	Leverage Buyout	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
(28)	Write-off	0.03	0.00	0.01	0.00	0.02	0.04	0.01	0.02	0.06	0.04	0.00	0.04	<u>0.07</u>	0.02	0.04	0.01	0.04	<u>0.10</u>	<u>0.08</u>	0.06	0.05	0.00	0.02	<u>0.28</u>	0.06	0.03	0.04

Table 4. Comparison of Proportions, Means, and Median for PSFs, Funds Affiliated to PSFs, and Funds Not Affiliated to PSFs

This table presents comparison of proportions tests (for seed, early stage, and expansion investments, and IPO, acquisition, LBO and write-off exits), and means tests and median tests (for industry market/book, staging and syndication) for PSFs, funds affiliated to PSFs, and funds not affiliated to PSFs. *, **, *** Statistically significant at the 10%, 5% and 1% levels, respectively.

	(1) PSF			(2) PSF Affiliated			(3) Non-PSF			Difference Tests (1) versus (2)		Difference Tests (1) versus (3)		Difference Tests (2) versus (3)	
	# Investees	Proportion or Mean	Median	# Investees	Proportion or Mean	Median	# Investees	Proportion or Mean	Median	Proportion or Mean	Median	Proportion or Mean	Median	Proportion or Mean	Median
Seed	26	0.81		45	0.42		904	0.14		3.34***		7.35***		3.83	
Early	26	0.15		45	0.27		904	0.10		-1.10		0.74		2.57	
Expansion	26	0.15		45	0.42		904	0.46		-2.28**		-3.21***		-0.56	
Industry Market / Book	26	4.28	4.74	45	5.45	4.38	904	4.56	3.94	-3.10***	p <= 0.381	-2.11**	p <= 0.001***	-2.11	p <= 0.008***
Staging	26	2.42	2.00	45	3.91	3.00	904	1.75	1.00	-2.29**	p <= 0.005**	1.34	p <= 0.000***	1.34	p <= 0.000***
Syndication	26	1.92	3.00	45	2.64	2.00	904	1.58	1.00	-2.14**	p <= 0.054**	1.40	p <= 0.000***	1.40	p <= 0.000***
IPO	26	0.00		45	0.09		787	0.11		-1.27		-1.80*		-0.54	
Acquisition	26	0.04		45	0.05		787	0.04		-0.12		0.19		0.43	
LBO	26	0.00		45	0.00		787	0.07		NA		-1.27		-1.66*	
Write-off	26	0.00		45	0.03		787	0.02		-0.61		-0.74		0.03	

Table 5. Logit Regression Analyses of Stage of Development

This table presents logit regression analyses of the probability of seed, early, expansion, late and buyout stage investments. Explanatory variables include the type of venture capital fund, MSCI market conditions (returns) in the year of investment, a dummy variable equal to one for the bubble years of 1999 and 2000, and the investment year (to account for trends over time). The numbers indicated are not the traditional logit coefficients; rather, the marginal effects are presented to illustrate the economic significance (the probability of the dependent variable being observed as a direct result of the explanatory variable). Two specifications are presented to illustrate robustness and the effect of two rounds of the IIF program. Models (7) - (10) exclude variables where there were no investments in the stage by the type of fund excluded as an explanatory variable, to avoid perfect collinearity. White's (1980) heteroskedasticity consistent covariance matrix estimator is used. Certain variables are excluded due to perfect collinearity problems (e.g., there were no late stage and buyout investments from PSFs). *, **, *** Significant at the 10%, 5% and 1% levels, respectively.

	Seed Stage	Early Stage	Expansion Stage	Late Stage	Buyout Stage
	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)
Constant	-5.628	-0.287	0.337	0.746	-9.601**
<u>Type of Fund</u>					
PSF	0.612***	0.070	-0.290***		
PSF Affiliate	0.045	0.031	-0.081		
IIF	0.452***	0.296***	-0.060	0.076*	-0.052**
IIF Affiliate	0.081	0.097*	0.313***	-0.009	-0.013
Public	0.077	-0.009	0.279***		-0.087***
Government	0.565***	0.237***	-0.021		
University	0.756***	0.042	-0.229*		
Green	0.019	0.204***	0.309***		
Corporate	0.413***	0.292**	0.045		-0.070**
Investment Bank	-0.058**	0.070**	0.306***	-2.937E-04	0.106***
Financial Institution	-0.048*	0.087***	0.158***	-0.001	0.224***
Individual	-0.048	0.064	0.066		0.273**
<u>Market Conditions and Learning</u>					
MSCI Investment Year	0.070	-0.083	0.183*	0.014	0.160***
Bubble	-0.056	-0.015	-0.022	0.051	-0.098***
Investment Year	0.003	1.846E-05	-2.756E-04	-0.0004	0.005**
<u>Model Diagnostics</u>					
Number of Observations	961	961	961	961	961
Number of Observations where Dependent Variable = 1	165	109	432	18	137
Loglikelihood	-311.128	-297.812	-597.074	-81.521	-329.977
Pseudo R ²	0.294	0.124	0.097	0.088	0.162
Chi-Squared Statistic	259.107***	84.022***	128.283***	15.813**	127.265***

Table 6. Logit and OLS Regression Analyses of Industry

This table presents logit and OLS regression analyses of the probability of investees in different industries being financed. Explanatory variables include the type of venture capital fund, MSCI market conditions (returns) in the year of investment, a dummy variable equal to one for the bubble years of 1999 and 2000, and the investment year (to account for trends over time). OLS is used where the industry market / book is the explanatory variable. Logit regressions are used for the industries indicated. The numbers indicated are not the traditional logit coefficients; rather, the marginal effects are presented to illustrate the economic significance (the probability of the dependent variable being observed as a direct result of the explanatory variable). Two specifications are presented to illustrate robustness and the effect of two rounds of the IIF program. Models (7) - (10) exclude variables where there were no investments in the stage by the type of fund excluded as an explanatory variable, to avoid perfect collinearity. White's (1980) heteroskedasticity consistent covariance matrix estimator is used. *, **, *** Significant at the 10%, 5% and 1% levels, respectively.

	Biotech / Medical	Computer	Internet	Communications	Industry Market / Book
	Model (6)	Model (7)	Model (8)	Model (9)	Model (10)
Constant	-5.030	4.722	0.959	2.581	85.305**
<u>Type of Fund</u>					
PSF	0.191*	-0.007		0.110	-0.296
PSF Affiliate	0.070	0.037	0.036	-0.016	0.304
IIF	0.250***	0.142**	0.136**	0.014	1.071***
IIF Affiliate	-0.042	0.101*	0.127**	0.017	0.899***
Public	0.080*	-0.076***	0.110**	-0.031***	0.324
Government					1.176***
University		-0.014			0.240
Green		-0.049	0.042	-0.012	0.063
Corporate	0.064	0.233**	0.084		0.802
Investment Bank	0.003	-0.078***	0.160****	0.017	0.652***
Financial Institution	0.029	-0.022	0.063**	-0.025**	0.090
Individual		-0.052	0.043	0.051	-0.437
<u>Market Conditions and Learning</u>					
MSCI Investment Year	0.131**	-0.011	-0.098	0.038	-0.241
Bubble	0.004	0.138**	0.064	0.043	0.208
Investment Year	0.002	-0.002	-0.001	-0.0013	-0.041**
<u>Model Diagnostics</u>					
Number of Observations	961	961	961	961	961
Number of Observations where Dependent Variable = 1	132	156	109	43	NA
Loglikelihood	-363.307	-403.942	-312.485	-162.022	-1975.176
Pseudo (Adjusted for (10)) R ²	0.055	0.052	0.080	0.077	0.059
Chi-Squared Statistic (F-statistic for (10))	42.448***	44.552***	54.676***	27.185***	4.99***

Table 7. Poisson and Box Cox Regression Analyses of Staging, Syndication, Portfolio Size, and Location

This table presents Poisson regression analyses of staging and syndication, Box Cox regression analyses of portfolio size per manager, and logit regression analyses of the probability that the investor and investee reside in the same state. Explanatory variables include the type of venture capital fund, MSCI market conditions (returns) in the year of investment (for staging and syndication), MSCI returns from the year of first investment to the last investment (or to 2005(Q4) for funds not yet fully invested), a dummy variable equal to one for the bubble years of 1999 and 2000, and the investment year (to account for trends over time). Dummy variables for investee characteristics are used for models (11) - (12) and (14), and the proportion of investees in the portfolio with the characteristic are used in model (13). The numbers indicated for (11) - (12) and (14) are the marginal effects, and not the traditional Poisson coefficients; the marginal effects are presented to illustrate the economic significance (the probability of the dependent variable being observed as a direct result of the explanatory variable). Coefficients and elasticities are indicated for the Box Cox models (13). White's (1980) heteroskedasticity consistent covariance matrix estimator is used. *, **, *** Significant at the 10%, 5% and 1% levels, respectively.

	Staging	Syndication	Portfolio Size / Manager		Investor & Investee Same State
	Model (11)	Model (12)	Model (13)		Model (14)
			Coefficient	Elasticity	
Constant	2.283	133.609***	-1.369***	-0.507	-3.628
<u>Type of Fund</u>					
PSF	0.359	0.318	-0.937***	-0.004	-0.240**
PSF Affiliate	0.298	0.504**	-0.152	-0.002	0.052
IIF	1.137***	1.111***	-0.360***	-0.007	0.061
IIF Affiliate	0.534***	0.713***	-0.349***	-0.011	-0.079
Public	0.691***	-0.119			-0.145**
Government	0.082	-0.151			0.061
University	0.498	0.527			0.042
Green	0.215	0.054			-0.128**
Corporate	0.677*	0.739			0.222*
Investment Bank	0.711***	0.077			-0.158***
Financial Institution	0.547***	-0.259			-0.173***
Individual		-0.010			
Fund Size			0.254***		
<u>Market Conditions and Learning</u>					
MSCI Investment Year (Investment Horizon for (13))	-0.536	-1.446***	2.435***	0.114	0.072
Bubble	-0.072	-0.016		0.905	0.002
Investment Year	-0.001	-0.067***		0.225	
Investment Duration			0.547***		
<u>Investee Characteristics</u>					
Seed	-0.108	0.692***	1.397***	0.520	0.036
Early	0.153	1.051***	0.339	0.126	-0.162***
Expansion	-0.070	1.038***	0.114	0.043	-0.001
Late		0.924***	-1.949***	-0.723	
Buyout		1.007***	0.240	0.089	
Domestic	-0.114	-0.304	0.204	0.077	
Industry Market / Book	-0.013	0.029	1.276***	0.513	-0.011
Total \$ Investment	0.003***	0.002	-0.115	-0.064	-0.0001
<u>Box Cox Curvature Parameter</u>					
Lambda	NA	NA	0.046	NA	NA
<u>Model Diagnostics</u>					
Number of Observations	852	852	164		763
Loglikelihood	-1178.825	-1308.982	-175.056		-504.591
Pseudo (Adjusted for (13)) R ²	0.063	0.010	0.791		0.046
Chi-Squared Statistic (F-statistic for (13))	159.792***	2.210	37.14***		48.338***